

A principle of balancing non-alternation and alternation in tonal metres

Abstract: Poetic metres can govern the basic rhythmic patterns of verses and lines, and generally require the phonetic attributes they regulate (e.g. stress, weight, and/or tone) to alternate within a line to avoid monotony. This paper proposes a new principle, namely that metre can balance both alternation and non-alternation within a line, using Middle Chinese regulated verse as an example. I argue that the canonical metrical templates of regulated verse require tones to alternate within a line, but only to a restricted extent. Beyond analysing the surface templates, the paper provides further support by showing that the *guping* (孤平) phenomenon in regulated verse can be analysed in Optimality Theory using constraints that encode alternation and non-alternation. Finally, I show that this balancing property of regulated verse metre not only enriches the crosslinguistic typology of poetic metres, but also illustrates how metres can be informed by – and thus serves as a window into – natural language phonology.

Keywords: metrics, tonal languages, *guping*, Chinese poetry, regulated verse

Poetic metres regulate the format of poetry in various aspects including prosody, rhyme and antithesis. From the prosodic perspective, metres govern the basic rhythmic patterns of verses and/or lines in verses. The metres of Indo-European languages such as English, French, Greek, and Latin have been intensively studied by generative phonologists (e.g. Fabb & Halle, 2008). Some metres even serve as a primary data source for the study of a language's phonology, especially when the language itself is now extinct (e.g. Sanskrit). In contrast, metres for poetry in tonal languages are relatively unexplored in the modern generative framework. Poems written in Chinese at most stages of its development employ complex metres as well. Although these metres have been described extensively in traditional Chinese literature and briefly analysed in several papers within the generative framework (e.g. Chen, 1979; Graham, 1980; Lorentz, 1980; Yip, 1980; Xue, 1989; Napoli, 1989, 1991), they have not been compared with the phonology of Chinese or other tonal languages.

In this paper, I analyse the metre of 'regulated verse' in Middle Chinese and propose that it calls for a new principle in generative metrics, namely the coexistence of non-alternation and alternation in the feature of a language modulated by the metre. Metrical templates for regulated verse dictate that tones must alternate in a line according to their categories, but not too abruptly. In addition, I argue that this feature of the metres of regulated verse mirrors the general tonal preferences in Chinese phonology.

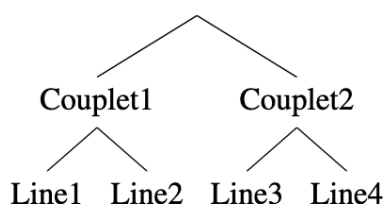
The paper is arranged as follows. Section 1 introduces regulated verse and its metrical templates. Section 2 makes observations of the templates and explains why it can be concluded that the regulated verse metre requires both alternation and non-alternation of tones. Section 3 provides a case study on a rule in regulated verse called ‘No Lone Even’ to demonstrate the effects of the interaction between alternation and non-alternation in Optimality Theory. Section 4 discusses how the metre of regulated verse is informed by and in turn informs natural language phonology. Section 5 concludes.

1. Background¹

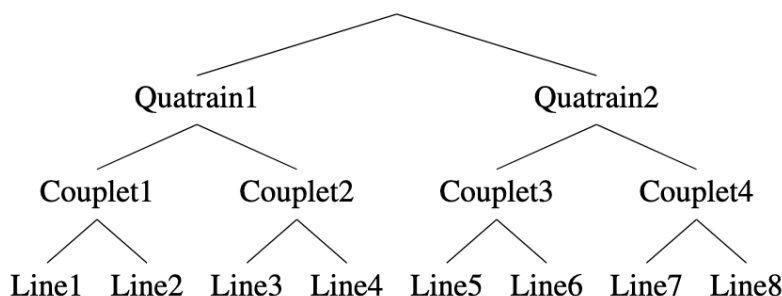
Regulated verse (近体诗) is a type of traditional Chinese poetry flourished in the Tang Dynasty (618–907 AD). Poems generally have four or eight lines: four-line poems are called ‘jueju’ (绝句) and eight-line poems are called ‘lushi’ (律诗). Each line has five or seven characters/syllables. Every two lines make up a couplet (联), and every four lines make up a quatrain (1).

(1) How lines are grouped into couplets and quatrains in regulated verse

a. Four lines:



b. Eight lines:



Middle Chinese was spoken at the time regulated verse became popular. There were four tones in Middle Chinese, which were divided into two categories. The level (píng 平) tone constitutes its own category ‘Even’ (píng shēng 平声), and the rising (shǎng 上), falling (or ‘departing’, qù 去) and checked (or ‘entering’, rù 入) tones make up the category ‘Oblique’ (zè shēng 仄声). I will represent these two categories with E and O respectively in what follows.

¹ Unless otherwise stated, the descriptions of Chinese poetry come from Wang (1957, 1962) and 冯胜利 (2015).

Poets were required to strictly follow the rigid metrical templates that restrict the tone category allowed to appear at each position in a line. Such regular alternation of tone sequences exists not only within the line but also between two adjacent lines in a couplet. The canonical templates, arranged by the number of syllables in a line and the tone of the first character in a poem, are given in (2)². Depending on the number of syllables in a line (5 or 7) and the tone category to which the first character of the poem belongs (even or oblique), there are four templates in total for four-line poems. Four couplet types are defined in the same way. Heptasyllabic lines are extensions of pentasyllabic lines by adding two syllables carrying the opposite tone to that of the first (two) syllable(s) at the beginning of the latter. For example, the first line of Couplet-7E, **E**EOO**E**EO, is constructed by adding two E-toned syllables at the beginning of OO**E**EO, the first line of Couplet-5O. Hence, I will focus primarily on the last five syllables of each line.

(2) Canonical templates for regulated verse

		initial tone					
		even		oblique			
length	five	a.	Couplet-5E	EEEEOO OOOEE	b.	Couplet-5O	OOEEEO EEOOE
			Couplet-5O	OOEEEO EEOOE		Couplet-5E	EEEEOO OOOEE
	seven	c.	Couplet-7E	EEOOEEEO OOEEEOOE	d.	Couplet-7O	OOEEEOOO EEOOOEE
			Couplet-7O	OOEEEOOO EEOOOEE		Couplet-7E	EEOOEEEO OOEEEOOE

Like many other metres, the templates for regulated verse allow some level of freedom. Free positions that are allowed to use the tone category other than the canonical one specified by the template are highlighted in bold in (2)³. The last position of each line is never free, which

² The templates for eight-line poems are always composed by doubling the template for the corresponding four-line poems with the same initial tone and line length, so this paper looks at four-line poems only.

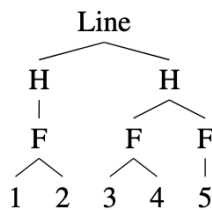
³ There has long been a misunderstanding that the free positions are positions 1 and 3 of pentasyllabic lines, and positions 1, 3, and 5 of heptasyllabic lines, as summarised by the ‘1-3-5’ Rule: ‘Licence for 1, 3, and 5, strictness for 2, 4, and 6’ (‘一三五不论，二四六分明’; 王力, 1957). However, the 1-3-5 Rule is in fact a misnomer for the actual generalisation; actually, only the first position of pentasyllabic lines and the first and third positions of heptasyllabic lines consistently allow inversions (王力, 1962; see more details on the exceptions to this rule in Section 3). Why the 1-3-5 Rule is problematic falls outside the scope of the current paper, so I will not discuss it in detail and simply follow the templates and generalisations outlined in (2).

is also statistically confirmed by Duanmu & Stiennon (2005: 24-25). Analogous to the tradition of Western poetic metrics, I will call the use of the non-canonical tone category ‘inversions’.

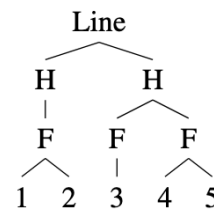
The templates for regulated verse have been analysed multiple times using linear (王力, 1957, 1962; Downer & Graham, 1963; T’sou, 1968; Jakobson, 1970; Lorentz, 1980; Napoli, 1991), hierarchical (Chen 1979; Graham, 1980; Yip, 1980; Xue, 1989) and grid-based (Napoli, 1989) approaches. Although these analyses differ significantly in their treatment of a line, most studies since Chen (1979) have agreed on two points. First, every line can be further divided into two half-lines or hemistiches (H) and every hemistich into feet (F). Second, the two syllables in each foot must carry tones from the same category (i.e. both E or both O). So, the pentasyllabic and heptasyllabic lines have the structures in (3) respectively. These analyses will serve as the basis of the observations and proposals put forward by this paper.

(3) Subline structure by Chen (1979)

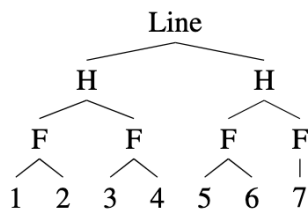
a. OOEEEO and EEOOE lines:



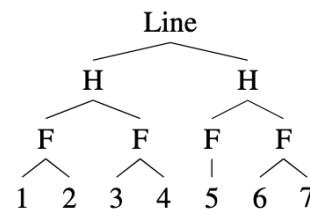
b. OOOEE and EEEEO lines:



c. OOEEEOE and EEOOEEO lines:



d. OOOEEEO and EEOOEEO lines:



(adapted from Chen, 1979:380)

2. Observations

Given the basics of the metrical templates for regulated verse, this section proposes that the tones of the syllables in each line need to alternate as well as not alternate at the same time.

2.1 Alternation

The feature that most straightforwardly illustrates the need for alternation in regulated verse is the cap on the number of consecutive syllables from the same tone category. Because the two syllables in every foot come from the same tone category, it is common to have multiple characters from the same category in a row within a line. However, the number of consecutive syllables from the same tone category allowed by the optimal metrical templates never exceeds three. For example, in Couplet-5E, one gets three syllables with E in succession at the beginning of the first line and three syllables with O in succession in the second line. The same

is observed for positions 3–5 of the two lines in Couplet-70. The absence of an excessive number of syllables from the same tone category shows that the metre regulates tone distribution within a line to avoid monotony, and it is important to have some degree of tonal variation within a line.

The other aspect that demonstrates the necessity of alternation is more subtle and subject to different analyses. Just like other accentual and quantitative metres, the metrical templates are believed to have positions alternating between strong and weak positions in terms of prominence. However, unlike other metres that are able to manipulate their natural phonological asymmetry such as stress/non-stress and heavy/light, it is difficult to tell which position is stronger in a foot in Chinese poetry, especially given that the two syllables within a foot must bear tones from the same tone category.

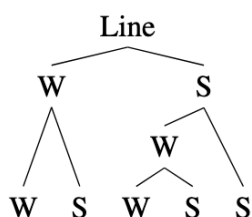
Therefore, phonologists have diverged in their analyses of relative prominence between positions. Chen (1979: 396) and Yip (1980: 108) both argued for an iambic foot because the recitation pattern of the verse shows a clear iambic rhythm. But Napoli (1991: 254-257) disagreed and argued that the recitation rhythm could not directly explain the tonal patterns of the templates, because evidence showed that the chanting rhythm is independent of the tones in Chinese verses in general (e.g. Boyce, 1980).

Despite these disagreements, analysts agree that there is a correlation between free positions (i.e. those that allow inversions) and the relative prominence of positions. In the arboreal analysis by Chen (1979) and Yip (1980), positions that allow the use of the opposite tonal category must be weak positions, which are the odd-numbered positions given the iambic feet in each line (4). In the grid analysis by Napoli (1989), the free positions tend to be those receiving a place on the second level of the grid (5), which are the more prominent positions in her analysis. In other words, the fact that free positions in a line (1 and 3) are not adjacent is the result of prominence alternation between positions in regulated verse.⁴

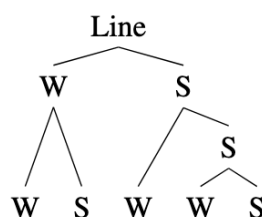
(4) Arboreal analysis (adapted from Yip, 1980: 115–116)

⁴ 端木三 (2016) also analysed many lines in regulated verse in with alternating weak and strong positions. But I decided not to intensively discuss his analysis here because the templates he provided were primarily based on the recitation stress pattern of the line and the number of morae in a syllable instead of tones.

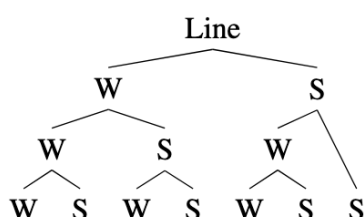
a. OOOEO and EEOOE lines:



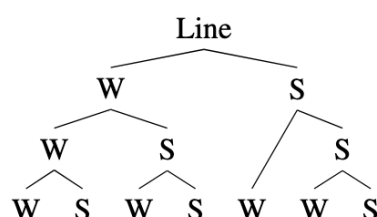
b. OOOEE and EEEEO lines:



c. OOEEOOE and EEOOEEO lines:



d. OOEEOEO and EEOOEEO lines:



(5) Grid analysis (adapted from Napoli, 1989: 28–29)

a. EEOOE lines:

x x / x x x

b. OOOEO lines:

x x / x x x

c. EEEEO lines:

x x / x x x

d. OOOEE lines:

x x x / x x x

e. OOEEOOE lines:

x x / x x x

f. EEOOEEO lines:

x x / x x x

g. OOEEOEO lines:

x x / x x x

h. EEOOEEO lines:

x x / x x x

In brief, alternation of tones is an important component of the tonal metrical templates for regulated verse. On the one hand, it is not allowed to have an excess number of tones in each line on the surface. On the other hand, the positions in each line seem to be asymmetrical to some extent, either in their recitation rhythms or in their ability to tolerate tones from the opposite category specified by the template.

2.2 Non-alternation

Although some degree of tone variation is necessary in regulated verse, perhaps in contrast with all other metres discussed in previous literature, the optimal metrical templates for regulated verse require that the tonal categories do not alternate too much at the same time.

Accentual metres stipulate that adjacent positions have different stress profiles, and quantitative metres stipulate that adjacent syllables differ in weight. However, this is not true for the metres of regulated verse. The templates in (2) show that, except the final position of each line, every position has at least one neighbouring position with a character from the same

tone category. That is, if a position has an E-tone syllable, at least one of the adjacent positions must also have an E-tone syllable, and the same applies to O-tone syllables. In other words, it is not possible to find ‘EOE’ or ‘OEO’ in the first four or six positions of each line in the optimal templates. Indeed, inversions could create ‘EOE’ or ‘OEO’ sequences: for example, using an E-toned syllable in the first position of an OOEE line leads to EOEE. But even taking inversions into account, a syllable flanked by two from the opposite tone category is not found in the last three positions (i.e. the second hemistich) of each line.

This pattern is in sharp contrast with accentual and quantitative metres, and even other tonal metres. For example, a metre that is greatly influenced by and thus very similar to Chinese regulated verse is the Vietnamese Luc Bat ‘six-eight’ stanza (6). Luc Bat also regulates the tone category allowed at each position of the line: flat words have either no tone or a low falling tone, and sharp words have tones with a high component. The metre for Luc Bat dictates that even-numbered positions must use words from the tone category specified, but odd-numbered positions can use words from any tone category, which are marked as ‘unregulated’ in (6). Unlike the regulated verse, the unregulated positions have no preference for tone category and it is possible to find a flat tone surrounded by two sharp tones or vice versa, even in the second half of the line.

(6) Vietnamese Luc Bat metres (adapted from Kiparsky 2020: 32)

×	♭	×	♯	×	♭ _a			
×	♭	×	♯	×	♭ _a	♯	♭ _b	
×	♭	×	♯	×	♭ _b			
×	♭	×	♯	×	♭ _b	♯	♭ _c	

(× = unregulated, ♭ = flat, ♯ = sharp, subscripts co-index required rhymes)

In short, although regulated verse does require the tones to alternate within a line, such alternation cannot be too vigorous, which can be demonstrated by requirement for each syllable to have at least one of its neighbours from the same tone category. Hence, I propose that the metres for regulated verse need alternation and non-alternation at the same time. To strengthen the argument, I present a case study that results from this requirement for simultaneous alternation and non-alternation in the next section.

3. Case Study: ‘Lone Even’

In traditional literature on Chinese poetry, there is an additional pattern that poets were required to avoid besides following the metrical templates outlined in (2) called ‘guping’ (孤平). This section explains what *guping* is and shows why it is a case demonstrating alternation and non-alternation at the same time when analysed with Optimality Theory (OT; Prince & Smolensky, 1993). Section 3.1 gives background information on *guping*. Section 3.2 briefly introduces how Optimality Theory works. Section 3.3 shows how one of the remedies to *guping*

can be accounted for using constraints representing alternation and non-alternation in Optimality Theory.

3.1 Background on *guping*

One can conclude from (2) that only the first position in pentasyllabic lines, and the first and third position in heptasyllabic lines can allow inversions. However, this does not apply to all the lines. The second line of Couplet-5O does not allow inversion in its first position, and the second line of Couplet-7E does not allow inversion in its third position, which is unsurprising given it is an extension to the second line of Couplet-5O.

Traditional Chinese poets and scholars believe that such a ban on inversions is intended to avoid a phenomenon called ‘guping’ 孤平 (王力, 1957), or literally ‘Lone Even’ (Duanmu & Stiennon, 2005: 5). Should these positions allow inversions, each of these two lines would end up with only one character from the E tone category (except the last position), as shown in (7a) where the lone E is underlined. However, using a character from the opposite tone category specified by the template in these positions is sometimes inevitable. For instance, the poet might be writing about a person or a city that does not have an alternative name. Under such circumstances, there are two ways of salvaging the line. One could save the line by changing the tone category of the character immediately after this lone E (in boldface), as shown in (7b). Alternatively, one could change the tone category of the character at the same position in the adjacent line of the same couplet (in boldface). That is, if a line’s tonal pattern ends up being OE^uOOE, it is allowed to change the other line in Couplet-5O into EO^uEEO; if a line ends up being OOOE^uOOE, it can be saved if the other line in Couplet-7E is EEE^uOEEO, as shown in (7c). Since Remedy 2 involves more lines and mechanisms, this paper will focus on Remedy 1, which – as I will show in Section 3.3 – is already sufficient in showing the balance between alternation and non-alternation in regulated verse metre.

(7) Hypothetical lines with ‘Lone Even’ and one remedy

	a. with ‘Lone Even’	b. Remedy 1	c. Remedy 2
Couplet-5O	O <u>E</u> OOE	O <u>E</u> EOE	EO ^u EEO O <u>E</u> OOE
Couplet-7E	OOO <u>E</u> OOE	OOO <u>E</u> EOE	EEE ^u OEEO OOO <u>E</u> OOE

It is fine to have this rule of ‘No Lone Even’ and its remedy as an addition to the metrical templates for regulated verse, but such rules have little explanatory power, and do not offer much insight as to why these positions are prohibited from inversions or why it is the E tone that cannot be ‘lone’ in a line. Instead, I argue that the solution to Lone Even in a line can be

derived naturally from the requirement of simultaneous alternation and non-alternation of regulated verse, which can in turn be represented by constraints in OT.

3.2 Basics of OT

OT is the theoretical framework initiated by Prince & Smolensky (1993) that relates the underlying representations (UR) of phonological forms to their surface forms (SF) using universal violable constraints, and the differences among languages arise from the differences in the rankings of these constraints. The constraints largely fall into two types: markedness constraints identify forbidden structures on the surface, and faithfulness constraints forbid mismatches between URs and SFs.

Potential surface realisations ('candidates') are evaluated for optimality based on the number of violations they incur and the constraint hierarchy. For example, in a constraint hierarchy $C1 \gg C2 \gg C3$ (where $\gg$ means 'dominates'), satisfaction of $C1$ is categorically more important than satisfaction of $C2$, which is in turn more important than satisfaction of $C3$ (McCarthy, 2007). To evaluate the relative optimality for every two candidates *cand1* and *cand2*, one needs to find the most highly ranked constraint C that differentiates them; if *cand1* violates C fewer times than *cand2*, *cand1* is more optimal than *cand2*. After comparing all the possible candidates for a UR in a pairwise fashion, the most optimal one is selected as the winner by the grammar consisting of the proposed constraints.

The tableaux⁵ in (8) include two minimal examples demonstrating how OT works. The highest constraint that differentiates *cand2* from *cand1* is $C1$, because *cand2* incurs one violation of $C1$ whereas *cand1* does not. So, *cand1* is more optimal than *cand2* and thus becomes the final winner predicted by the mini grammar in (8a). Sometimes, two candidates violate the same constraint by different times, then the one incurring fewer violations wins. Therefore, in (8b), *cand2* loses to *cand1* because the former violates $C2$ twice whereas the latter violates $C2$ just once.

(8) Examples of mini grammar in OT

a.

UR	C1	C2
☞ cand1		*
cand2	*!	

b.

UR	C1	C2
☞ cand1		*
cand2		**!

Finally, although constraints should eventually be completely ranked with respect to one another according to Prince & Smolensky's (1993) proposal, the available data in a language usually cannot determine the total ranking of the constraints. In practice, it is likely that some

⁵ The tableau notations in this paper follow the traditions in Prince & Smolensky (1993). ☞ = winning candidate; * = violation; ! = fatal violation (i.e. this violation disqualifies the candidate for further optimality comparisons).

constraints has not been relatively ranked with each other by the time the winning candidate is selected. For example, because C1 outranks C3, cand3 which violates C1 once loses to cand1 which only violates C3 once; similarly, cand2 also loses to cand1 because C2 outranks C3. Because cand1 wins over both cand2 and cand3 and there are only three contenders for optimality in the grammar in (9), cand1 is the most optimal candidate and the relative ranking between C1 and C2 is undetermined (indicated by dashed column lines).

(9) Unranked constraints

UR	C1	C2	C3
☞ cand1			*
cand2		*!	
cand3	*!		

In the next section, I will show how these basics of OT can be applied to the generalisations of regulated verse metre summarised earlier and how the remedy to a line with a Lone Even tone is a result of the interacting constraints representing the alternation and non-alternation nature of regulated verse.

3.3 Analysing *guping* in OT

How does one determine the URs and SFs in poetic metrics then? If one understands URs as the canonical and abstract representation of a morphophonemic form and SFs the deviations from the ‘ideal’ form in natural language, then the requirement of metres can be understood as the UR and the actual line pattern can be interpreted as SFs analogously. Therefore, in the case of regulated verse, the optimal metrical templates in (2) could be regarded as the UR, and the possible variants of the optimal template are the candidates for the SF. Among these variants, the acceptable one on the surface for a certain line is the winning candidate. The line with Lone Even could also be considered as a variant of the optimal line structure which unfortunately loses to the acceptable proposed remedy. So, what is needed is a set of constraints that can successfully map the optimal line shape to the remedy and avoid the candidate with the lone E.

Such constraints are deeply rooted in the generalisations made about the lines of regulated verse. To start with, the fact that every syllable needs at least one neighbouring character from the same tone category can be rephrased as a binarity requirement for tones. Constraints such as BIN-E which penalises an E tone flanked by two O tones (8a), and BIN-O penalising an O tone flanked by two E tones (8b), can capture such binarity. That only odd-numbered positions can allow inversions may be formulated as a constraint such as ODD-INVER (8c). Finally, the restriction that the last position of each line never permits inversions can be formalised as NO-INVER-FIN (8d).

(8) Necessary constraints and what they penalise

Constraints	Assign one violation for ...
a. BIN-E	each non-final E tone being flanked by two O tones.
b. BIN-O	each non-final O tone being flanked by two E tones.
c. ODD-INVER	each inversion occurring in an even-numbered position.
d. NO-INVER-FIN	each inversion occurring in the final position of a line.

The possible variants of the two lines with the prospect of having a Lone Even tone – the second line of Couplet-5O and the second line of Couplet-7E – are listed in (9) and (10) respectively⁶. The critical ranking to select the right candidate, namely the solution to Lone Even, is shown in the tableaux in these two examples: BIN-O needs to be outranked by BIN-E, ODD-INVER and NO-INVER-FIN.

(9) Couplet-5O

	EEOOE	BIN-E	ODD-INVER	NO-INVER-FIN	BIN-O
a. 	OEEOE				*
b.	OEOOE	*!			
c.	OOOOE		*!		
d.	OEOEE	*!	*		*
e.	OEEEO			*!	

(10) Couplet-7E

	OOEEOOE	BIN-E	ODD-INVER	NO-INVER-FIN	BIN-O
a. 	OOOEEOE				*
b.	OOOEOOE	*!			
c.	OOOOOOE		*!		
d.	OOOEOEE	*!	*		*
e.	OOOEEEO			*!	

Each of these constraints contributes to one aspect of the prohibition of Lone Even and its solution. Lone Even in a line implies an E-toned syllable surrounded by two syllables with O tones, so the binarity constraint BIN-E stops such lines from appearing on the surface (9b, 10b).

⁶ In the standard practice of OT, the fully faithful candidate should be included in the tableau as a potential candidate – in this case, it should be the optimal line shape in the canonical template. But given that this optimal line is always going to win over any other line with changes, and that the aim of the analysis is to find the best alternative when the most optimal line pattern cannot be obtained, I choose to omit the fully faithful line pattern in all the tableaux in this paper.

ODD-INVER ensures that, even if an E ends up having no neighbour from the same category, remedies should be sought in odd-numbered positions only. Thus, the candidates that invert the only E tone to an O tone cannot be solutions (9c, 10c), because such inversions take place in even-numbered positions. Similarly, adding another E-toned character by inversion in position 4 (or position 6 of heptasyllabic lines) is not a viable solution either (9d, 10d).

In fact, ODD-INVER restricts the search for solutions to positions 1, 3, and 5 (and 7 for lines with seven syllables). Since the first position already involves an inversion that led to the problem of Lone Even, and the final positions do not allow any inversions as required by NO-INVER-FIN, the only possible solution is to have an inversion at position 3 in a pentasyllabic line, or position 5 in a heptasyllabic line. In other words, the solution lies in the position ‘immediately after the lone E’ (9a, 10a), which matches perfectly with the description in traditional literature.

However, allowing an inversion immediately after the lone E leads to the violation of another constraint. The E-toned syllable after inversion in position 5 and the line-final E form a sequence of EOE with the O-tone syllable in between, and such a sequence violates BIN-O. But the proposed remedies for Lone Even do contain such sequences. Thus, NO-INVER-FIN must rank above BIN-O to make sure that the EOE sequence remains unchanged at the end of a line. Under such ranking, no inversion can occur in the final position of a line even if the EOE sequence violates the binarity required by BIN-O. BIN-E and ODD-INVER must also rank above BIN-O, given that the latter is the only constraint the winning candidate violates. The fact that BIN-E $\gg$ BIN-O also somehow explains why only E tones cannot be lone in a line: the requirement for the binarity in E is stronger than that for the binarity in O.

In brief, the interaction between OT constraints regulating alternation (ODD-INVER) and non-alternation (BIN-E, BIN-O, and NO-INVER-FIN) successfully derives the solution to a line with only one E tone from the optimal metrical template. This is an example that effectively demonstrates why the tones in regulated verse need to both alternate and not alternate at the same time.

4. Discussion

The previous two sections have shown, through the patterns of the optimal metrical templates and the solutions to the unacceptable line configuration ‘Lone Even’, that Chinese regulated verse requires the tones in each line to both alternate and not alternate simultaneously. In this section, I discuss how Chinese regulated verse contributes to the study of metres and the knowledge of natural language phonology in general.

4.1 Metre typology

First, the metre of regulated verse greatly enriches the typology of metres. Previous studies of metres usually focused on Indo-European languages, and most metres analysed in detail

were syllabic, accentual, or quantitative. Syllabic metres generally only count the number of syllables in a line and do not mandate that adjacent syllables alternate in quality. But accentual and quantitative metres do need the syllable feature they regulate – that is, the stress and weight of the syllable – to vary: one cannot have an entire line of unstressed/stressed syllables, or a line of light/heavy syllables, without any variation. The tonal metre of Chinese regulated verse resembles accentual and quantitative metres in that the tones of the syllables in a line also need to vary to some extent. As shown in Section 2.1, the maximum number of syllables from the same tone category in a line that the metrical templates allow is three. Regulated verse enhances the claim that metres track and govern variation in poetry. Regulated verse also behaves like accentual and quantitative metres in the sense that there also seem to be some inherent asymmetries in its positions. The freedom to use the non-canonical tone required by the template differs significantly between odd- and even-numbered positions in regulated verse. This difference has been argued to be comparable to the prominence alternation in accentual and quantitative metres.

On the other hand, the metres of regulated verse have a significant difference from other previously studied metres too. On top of the need for variation, another feature shared by accentual and quantitative metres is that the stress or weight they regulate needs to be different in adjacent syllables. In accentual metres, adjacent syllables need to alternate in stress: an unstressed syllable is usually surrounded by two stressed ones, and vice versa. In quantitative metres, adjacent syllables need to alternate in weight: a light syllable is usually surrounded by two heavy ones, and vice versa. However, the tonal metre of Chinese regulated verse is completely the opposite, in that no syllable may be surrounded by two characters from a different tone category, as required by the canonical templates. Although regulated verse also requires tones to alternate, it prefers them to change gradually rather than abruptly.

The enrichment of typology is not restricted to metres for languages that lack tones. Even other tonal metres differ from Chinese regulated verse in terms of the tone categories appearing in adjacent positions. As shown in Section 2.2, although Vietnamese *Luc Bat* also regulates tones in even-numbered positions more strictly, it does not require syllables to have one neighbour from the same category. In other words, non-alternation is neither an absolute requirement nor an inherent property of tonal language metres. Therefore, regulated verse provides a novel angle for phonologists to consider when studying metres, namely that non-alternation must be balanced with alternation in tonal metres.

4.2 Natural language and metre

Second, the metres for regulated verse reflect many features of natural languages with tones. The fact that there are never more than three characters from the same tone category allowed in the optimal template for regulated verse is similar to the number restriction on tone-bearing units (TBU) in many tonal languages. For example, many Bantu languages exhibit a bounded

tone spreading process called ‘tone tripling’, where an high tone⁷ spreads to the following two TBUs within a word, but no further, as exemplified in (11) with Shona. This process is also found in Kanye Tswana (Creissels, 1998), Venda, Kalanga, Tsonga, Copperbelt Bemba (Odden & Marlo, 2019), and Dembwa Taita I (Odden, 2001). In Prinmi, a Tibeto-Burman language spoken in China, no more than two H-toned TBUs and no more than three L-toned TBUs are allowed in a word, with neither type exceeding three (Ding, 2006; Hyman, 2009; Jardine, 2020). (12) provides an example containing three consecutive L-toned TBUs and one containing two consecutive H-toned TBUs, with their respective tonal patterns marked on the right.

(11) Shona H tone tripling (Odden, 2015)

kutóra ‘to make’
kutóresá ‘to make take’
kutóreséra ‘to make take for’
kutóresérana ‘to make take for each other’

(12) Prinmi tone spreading restrictions (Ding, 2006: 14)

bíb'ob'oge ‘as for roasted flour’ HLLL
töpúm'zle ‘donkey tail’ LHHH

Another feature of the metres of regulated verse that resembles tones in natural languages is the prohibition of rapid excursions in tone (McPherson, 2016). In regulated verse, an E tone flanked by two O tones, or an O tone flanked by two E tones, is banned within a line, suggesting that the change of tones cannot be too abrupt. Similarly, many tonal languages around the world also prohibit a tone surrounded by two tones of a different type, captured by the *HLH (Cahill 2007) or the *TROUGH (Yip 2002:137) constraint. For example, in Kɔnni (Gur, Ghana), when a sequence of HLH would be expected to occur by combining morphemes, the second H spreads left, delinking the L and causing downstep (13). Other languages that disprefer HLH sequences include Deg, Ga, Esaaka Makhwa, and Zulu in Africa (Odden, 2000), Mianmin and Kairi in Papua New Guinea (Cahill, 2000), and Inapari (Arawakan, Peru) and Mamainde (Nambikwaran, Brazil) in South America (Parker, 1999; Eberhard, 2007).

(13) Kɔnni disprefers HLH sequences (Cahill, 2007)

a. púra ‘ chests’
b. pú'ráhá ‘the chests’
c. $\begin{array}{ccccccc} & H & & H & & H & L & H \\ & | & & | & & | & / & | \\ \text{p} & \text{u} & \text{r} & \text{a} & + & \text{h} & \text{a} & \rightarrow & \text{p} & \text{u} & \text{r} & \text{a} & \text{h} & \text{a} \end{array}$

⁷ In this paper, high tones are indicated with an acute accent. All other unmarked TBUs are low-toned.

Even Chinese has a similar phenomenon, where a dip in pitch is not allowed between two high tones in (Beijing) Mandarin (Chen, 2000). T2 sandhi stipulates that a non-final rising tone [35] becomes a high tone [55] when preceded by a rising [35] or high [55] tone (14), because each of the [55. 35] and [35. 35] sequences can create a dip in pitch within them. Similarly, the *Yi-bu-qi-ba* rule which only applies to yi 一 ‘one’, bu 不 ‘not’, qi 七 ‘seven’, and ba 八 ‘eight’ obligatorily changes the tone of these syllables into [35] when followed by another falling [53] tone syllable (15a).

(14) T2 sandhi (Chen, 2000: 21-22)

- | | |
|--------------------------------|--------------------------------------|
| a. 天文台 ‘observatory’ | b. 人民币 ‘Renminbi (Chinese currency)’ |
| 55. 35. 35 base form | 35. 35. 51 base form |
| 55. 55. 35 SF (with T2 sandhi) | 35. 55. 51 SF (with T2 sandhi) |

(15) Minimal pairs due to *Yi-bu-qi-ba* rule (Chang, 1992: 166f; Chen, 2000: 22)

- | | |
|--|-----------------------|
| a. 不对 ‘not correct’ | b. 部队 ‘troops’ |
| 53. 53 base form | 53. 53 base form |
| 35. 53 SF (with <i>Yi-bu-qi-ba</i> rule) | 53. 53 SF (no change) |

An interesting asymmetry between H and L tones exists within a single language as well as crosslinguistically: HLH is commonly banned whereas LHL occurs abundantly. Cahill (2007) attributed this asymmetry to the markedness of H tones, given that systems in which H tones are marked are more frequent than those in which L tones are marked (Maddieson, 1978). This asymmetry has a similar parallel in the metrical templates for regulated verse. On the one hand, there exists a rule that disallows a single E tone in a line, namely the ‘No Lone Even’ rule, but there is no corresponding rule for O tones. On the other hand, although both EOE and OEO are dispreferred, the latter incurs a worse violation due to the relative ranking BIN-E >> BIN-O.

In brief, several features of the metres of regulated verse have clear parallels in natural languages with tones, suggesting that metres are not random choices by poets but are, in fact, heavily informed by natural language phonology. More importantly, these features are not confined to Chinese but are widely observed in many tonal languages around the world. Therefore, the metrical characteristics of regulated verse are arguably a reflection of human preference for tonal patterns in general.

5. Conclusion

This paper proposes a new principle of balancing non-alternation and alternation in tonal metres with evidence from the metres of Middle Chinese regulated verse. Regulated verse requires the tones of syllables in each line to alternate in tone categories and not alternate too

abruptly at the same time. The alternation requirement is observed in the number of consecutive syllables allowed to carry tones from the same category, as well as in the asymmetry of the positions' ability to permit the use of the opposite tone category specified by the canonical template. The non-alternation requirement is found in the fact that each syllable must have at least one neighbour with a tone from the same category. The fact that these two requirements must be balanced can be demonstrated with a rule named 'No Lone Even' in regulated verse, which – when analysed with the tools of generative metrics – naturally falls out as a result of the interaction of the OT constraints representing alternation and non-alternation. This feature of regulated verse significantly enriches the typology of poetic metres.

It is also shown that these characteristics of regulated verse greatly resemble those of the phonology of tonal languages in general. The 'up-to-three' restriction on the number of consecutive tones in the optimal metrical templates is similar to tone tripling in many tonal languages around the world, and the dispreference for EOE and OEO sequences patterns with the prohibition of rapid reversals in pitch in tonal languages, including Mandarin Chinese. In conclusion, regulated verse is another case showing that metres are not merely a matter of poets' personal preference, but the most balanced option under the tension between the natural tendency of tones to spread and the rhythmic alternation required in poetry.

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声调格律系统中“非交替”与“交替”的平衡原则

摘要: 诗歌格律可以制约诗篇及诗行的基本节律格局,并通常要求其所规约的音系属性,如重音、音节重量或声调等,在诗行内部呈交替分布,以避免节律上的单调。本文以中古汉语近体诗的平仄格律为例,提出一条新的格律原则:诗行内部的格律组织并非单纯追求交替,而是需要在交替与非交替之间形成平衡。本文认为,近体诗的标准律式固然要求句内平仄相间,但这种交替要求只在受限范围内实现。除对表层格律模板的分析之外,本文进一步指出,近体诗中的“孤平”现象可以在优选论(Optimality Theory)框架下,通过表征交替与非交替的制约条件及其等级排列加以分析。最后,本文说明,近体诗格律的这种平衡性质不仅丰富了诗歌格律的跨语言类型学,同时也表明诗歌格律可以由自然语言音系提供理据,并由此成为观察自然语言音系的重要窗口。

关键字: 格律 声调语言 孤平 汉语诗歌 近体诗